Curriculum Vitae

Byung Hun Lee, Ph.D.

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Contact

Postdoctoral Researcher
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EDUCATION

Mar. 2015 ~ Aug. 2021	**Seoul National University** ***Ph.D**. in Physics* Dissertation: Imaging mRNA in live neurons and animals Advisor: Prof. Hye Yoon Park	Seoul, Republic of Korea
Mar. 2018 ~ Feb. 2021	Alternative military service as a research personnel (전문연구요원)	
Mar. 2011~ Feb. 2015	**Sungkyunkwan University** ***B.S.** in Physics*	Suwon, Republic of Korea

RESEARCH EXPERIENCE

Seoul National University	Department of Physics and Astronomy	Seoul, Korea
Graduate Student	Advisor: Prof. Hye Yoon Park	2015.03-2021.08
Postdoctoral Researcher	Mentor: Prof. Hye Yoon Park	2021.09-2022.02
Harvard University	Department of Chemistry and Chemical Biology	Cambridge, MA
Postdoctoral Researcher	Mentor: Prof. Adam E. Cohen	2022.03-Present

- **Axonal *β-actin* mRNA dynamics:** Initiated the project to investigate how β-actin mRNAs transported and localized in axons. I used transgenic mice in which β-actin mRNAs are labeled with dozens of GFPs (Park *et al*, *Science*, 2014) to visualize β-actin mRNA in live

neurons and investigated how the β-actin mRNAs are transported in axons. To obtain trajectories of mRNAs, I developed a tracking software, ***HybTrack***.

- **Real-time imaging of *Arc* mRNA synthesis in live mouse brain:** Using the PP7 mRNA tagging technique, for the first time, the transcription event of Arc, one of the immediate-early genes, was observed in live mice. Arc transcription events of hundreds of neurons in the hippocampal CA1 and the retrosplenial cortex (RSC) were identified during memory formation and retrievals. Besides, employing a virtual reality system, I investigated the relation between calcium activities and the Arc transcription.
- ***In vivo* dendritic voltage imaging:** Now as a postdoc researcher under supervision of Prof. Adam E. Cohen, I address a longstanding question: what is the computational significance of dendritic excitations *in vivo*? I perform high-speed, comprehensive voltage imaging across the dendritic trees of hippocampal CA1 pyramidal neurons in awake mice navigating in virtual reality.

PUBLICATIONS

Pre-print

1. **B. H. Lee**, P. Park, X. Wu, J. D. Wong-Campos, J. Xu, M. Xiong, Y. Qi, Y. Huang, D. G. Itkis, S. E. Plutkis, L. D. Lavis, A. E. Cohen (2025). Fast dendritic excitations primarily mediate back-propagation in CA1 pyramidal neurons during behavior. ***bioRxiv (under revision)***
2. X. Wu, **B. H. Lee**, P. Park, J. D. Wong-Campos, J. Xu, S. E. Plutkis, L. D. Lavis, A. E. Cohen (2025). A dendrite-resolved, in vivo transfer function from spike patterns to dendritic Ca2+**. *bioRxiv (under revision)***
3. D. G. Itkis, F. P. Brooks III, H. C. Davis, R. Hotter, D. J. Wong-Campos, Y. Qi, B. Z. Jia, M. Howell, M. Xiong, R. F. Hayward, **B. H. Lee**, Y. Wang, R. T. Perelman, A. E. Cohen (2025). Luminos: open-source software for bidirectional microscopy. ***bioRxiv***

Peer-reviewed primary publications: († Equal contributions)

1. J. D. Wong-Campos†, P. Park†, **B. H. Lee†**, H. C. Davis, Y. Qi, H. Tian, D. G. Itkis, D. Kim, J. B. Grimm, S. E. Plutkis, L. D. Lavis, A. E. Cohen (2026). Voltage dynamics of cortical dendrites *in vivo* ***Nature Neuroscience***
2. P. Park†, J. D. Wong-Campos†, D. G. Itkis, **B. H. Lee**, Y. Qi, H. C. Davis, B. Antin, A. Pasarkar, J. B. Grimm, S. E. Plutkis, K. L. Holland, L. Paninski, L. D. Lavis, A. E. Cohen (2025). Dendritic excitations govern back-propagation via a spike-rate accelerometer ***Nature Communinations 16,*** 1333
3. D.W. Kim, H. C. Moon, **B. H. Lee**, H. Y. Park (2024). Decoding Arc transcription: a live-cell study of stimulation patterns and transcriptional output. ***Learning & Memory Vol. 31***, No. 8, a054024.124
4. K. Noh, WH Cho., **B. H. Lee**, D. W. Kim, Y. S. Kim, K. Park, M. Hwang, E. Barcelon, Y.

K. Cho, C. J. Lee, B. Yoon, S. Choi, H. Y. Park, S. B. Jun, S. J. Lee (2023). Cortical astrocytes modulate dominance behavior in male mice by regulating synaptic excitatory and inhibitory balance. ***Nature Neuroscience 26***, 1541-1554

5. C. Lee†, **B. H. Lee**†, H. Jung†, C. Lee, Y. Sung, H. Kim, J. Kim, J. Y. Shim, J. Kim, D. I. Choi, H. Y. Park, B. -K. Kaang (2023). Hippocampal engram networks for fear memory recruit new synapses and modify pre-existing synapses in vivo. ***Current Biology 33,*** 507-516.
6. WH Cho., K. Noh, **B. H. Lee**, E. Barcelon, S.B. Jun, H. Y. Park, S. J. Lee (2022). Hippocampal astrocytes modulate anxiety-like behavior. ***Nature Communications*** **13**, 6536.
7. H. Choi, **B. H. Lee**, H. Y. Park (2022). Time-resolved analysis of transcription kinetics in single live mammalian cells. ***Front. Phys. 10:977125.***
8. **B. H. Lee**, S. Bang, S. Lee, N. L. Jeon, H. Y. Park (2022). Dynamics of axonal β-actin mRNA in live hippocampal neurons. ***Traffic*** **23**, 496-505.
9. **B. H. Lee**†, J. Y. Shim†, H. C. Moon, D. W. Kim, J. Kim, J. S. Yook, J. Kim, H. Y. Park (2022). Real-time visualization of mRNA synthesis during memory formation in live animals. ***Proc. Natl. Acad. Sci. U.S.A. 119***. **–** Featured in **preLights** (The Company of Biologists): link
10. Kim, Y., S. Kang, **B. H. Lee**, Y. J. Song, S. Kang, H. Y. Park and Y. Lee (2022). De novo generation of a bright blue fluorophore from 2-oxoglutarate in biological samples. ***Chemical Science 13*** (2): 365-372.
11. M. S. Song, H. Baek, K. Lee, D. Yoo, K. Chung, J. Lee, H. C. Moon, **B. H. Lee**, H. Y. Park, G. Yi (2021). Intracellular gallium nitride microrod laser, ***NPG Asia Materials 13.***
12. **B. H. Lee**, H. Y. Park (2018). HybTrack: A hybrid single particle tracking software using manual and automatic detection of dim signals. ***Scientific Reports*** **8**.

Review articles and Book chapters:

1. **B. H. Lee**, S. W. Bae, J. J. Shim, S. Y. Park, H. Y. Park (2016). Imaging Single-mRNA Localization and Translation in Live Neurons. ***Molecules and Cells*** **39**, 841-846 (2016).
2. H. C. Moon, **B. H. Lee**, K. Lim, J. S. Son, M. Song, H. Y. Park (2016). Tracking single mRNA molecules in live cells. ***Journal of Physics D: Applied Physics*** **49**.
3. J. Y. Shim, **B. H. Lee**, H. Y. Park (2019). Visualization of Single mRNAs in Live Neurons. ***Methods in Molecular Biology***, DOI: 10.1007/978-1-4939-9674-2_4.
4. H. Choi, **B. H. Lee**, H. Y. Park (2021). Quantitative models for transcriptional dynamics monitored using an MS2-GFP system. ***Journal of the Korean Physical Society 78***, 365-372, DOI: 10.1007/s40042-020-00037-3.

Fellowships

2023.9 - 2024.8	2023 Academic Research Support Program for Engineering and Science (Postdoctoral Overseas Training) from National Research Foundation of Korea (NRF)

AWARDS AND HONORS

- NEBS–GC Biopharma Award, New England Bioscience Society, Boston, USA, 2026
- Brain Korea Graduate student thesis award, 2021
- Brain Korea PhD candidate scholarship, 2019
- Best **oral** presentation awards in Korean Physics Society 2018, 2019
- Best **poster** presentation awards in Korean Physics Society 2016, 2017
- Brain Korea best **TA** awards 2015

SKILLS AND TECHNIQUES

- **Surgical Techniques**: Cranial window surgery (target regions: CA1, RSC, V1, mPFC; includes prism implantation), In utero electroporation, single-cell electroporation, perfusion
- **Fluorescence Imaging**: Two-photon, wide-field, and TIRF microscopy
- **Image Analysis**: Image processing (custom MATLAB scripts), functional recording analysis
- **Computational Tools**: MATLAB (github.com/bhlee1117), Blender (3D modeling & printing), Adobe Illustrator, Procreate, CAD (Fusion360/Onshape)
- **Molecular Biology**: DNA cloning, AAV and lentivirus packaging, single molecule fluorescence in situ hybridization (smFISH), immunostaining
- **Cell Biology**: Primary neuron culture

CONFERENCE PRESENTATIONS

POSTER:

1. **B. H. Lee**, S. Bang, S. Lee, N. L. Jeon, H. Y. Park. Axonal mRNA dynamics in live hippocampal neurons. 60th Annual Meeting of Biophysical Society. Feb. 27- Mar. 02, 2016, Los Angeles, CA.
2. **B. H. Lee**, S. Bang, S. Lee, N. L. Jeon, H. Y. Park. Axonal mRNA dynamics in live hippocampal neurons. 2016 Spring Meeting of Korean Physics Society. April 20-22 2016,

Daejeon, Korea

3. **B. H. Lee**, S. Bang, S. Lee, N. L. Jeon, H. Y. Park. Axonal mRNA dynamics in live hippocampal neurons. 2016 Neuroscience meeting (Society for Neuroscience). Nov. 12-16, 2017, San Diego, CA.
4. **B. H. Lee**, H. Y. Park (2018). HybTrack: A hybrid single particle tracking software using manual and automatic detection of dim signals. 2017 fall Meeting of Korean Physics Society. Oct. 25-27, 2017, Gyeongju, Korea.
5. **B. H. Lee**, S. Bang, S. Lee, N. L. Jeon, H. Y. Park. Axonal mRNA dynamics in live hippocampal neurons. 2017 Neuroscience meeting (Society for Neuroscience). Nov. 12-16, 2017, Washington, DC.
6. **B. H. Lee**, S. Bang, S. Lee, N. L. Jeon, H. Y. Park. Axonal mRNA dynamics in live hippocampal neurons. 2018 Neuroscience meeting (Society for Neuroscience). Nov. 3-7, 2018, San Diego, CA.
7. **B. H. Lee†**, J. Y. Shim†, H. C. Moon, D. W. Kim, J. Kim, J. S. Yook, J. Kim, H. Y. Park. Imaging activity-dependent transcription of endogenous Arc mRNA in live mouse brain. 2019 fall Meeting of Korean Physics Society. Oct. 23-25, 2019, Gwangju, Korea.
8. **B. H. Lee†**, J. Y. Shim†, H. C. Moon, D. W. Kim, J. Kim, J. S. Yook, J. Kim, H. Y. Park. Real-time visualization of mRNA synthesis during memory formation in live animals. **HHMI virtual science meeting, Cognitive and Systems Neuroscience**, October 12-13, 2021.
9. **B. H. Lee**, P. Park, D. J. Wong-Campos, A. E. Cohen. All-optical electrophysiology reveals rules of complex spike formation in hippocampus 7th Annual Neurophotonics Center Symposium, Jan. 17th, 2024, Boston, MA.
10. **B. H. Lee**, P. Park, X. Wu, D. J. Wong-Campos, A. E. Cohen. Mapping voltage dynamics of dendritic arbors in behaving mice. 2024 Neuroscience meeting (Society for Neuroscience). Oct. 5-9, 2024, Chicago, IL.
11. **B. H. Lee**, P. Park, X. Wu, D. J. Wong-Campos, A. E. Cohen. Mapping voltage dynamics of dendritic arbors in behaving mice. **Gordon Research Conference, Dendrites: Molecules, Structure and Function**, March 22-23, 2025, Ventura, CA, United States.

ORAL:

1. **B. H. Lee**, S. Bang, S. Lee, N. L. Jeon, H. Y. Park. Axonal mRNA dynamics in live hippocampal neurons. **2018 Spring Meeting of Korean Physics Society**, April 25-27 2018, Daejeon, Korea.
2. **B. H. Lee**, S. Bang, S. Lee, N. L. Jeon, H. Y. Park. Axonal mRNA dynamics in live hippocampal neurons. **2018 Fall Meeting of Korean Physics Society**, Oct. 24-26, 2018, Changwon, Korea.
3. **B. H. Lee†**, J. Y. Shim†, H. C. Moon, D. W. Kim, J. Kim, J. S. Yook, J. Kim, H. Y. Park. Imaging the dynamics of engram in CA1. **2019 Spring Meeting of Korean Physics**

Society, April 24-26 2019, Daejeon, Korea.

4. **B. H. Lee**†, J. Y. Shim†, H. C. Moon, D. W. Kim, J. Kim, J. S. Yook, J. Kim, H. Y. Park. Real-time visualization of mRNA synthesis during memory formation in live animals. **HHMI virtual science meeting, Cognitive and Systems Neuroscience**, October 12-13, 2021.
5. **B. H. Lee**, P. Park, X. Wu, D. J. Wong-Campos, A. E. Cohen. Mapping voltage dynamics of dendritic arbors in behaving mice. **Gordon Research Seminar, Dendrites: Molecules, Structure and Function**, March 22-23, 2025, Ventura, CA, United States.
6. **B. H. Lee,** P. Park, X. Wu, D. J. Wong-Campos, A. E. Cohen. Fast dendritic excitations primarily mediate back-propagation in CA1 pyramidal neurons during behavior, **New England Bioscience Society Annual Meeting**, May 23, 2026, Boston, MA, United States.